

Assignment 1 – Computer Vision (20% weightage)

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Submission Due: 20/3/2026 (16:00 pm)

Instructions:

1. All code should be in Python 3.11.x and use *Skimage* imaging library only. OpenCV or PIL cannot be used.
2. There should be one common *Report.doc* or *Report.pdf* for all the tasks below.
3. Submit a single zip file containing the task python files, report and images.
4. A penalty of 10% of the maximum possible mark will be deducted per 24 hours from time submission is due for up to 7 days. After 7 days, you will receive a mark of 0.
5. Submission is to be made only on Ultra (Learn.uq).
6. Plagiarism cases will be dealt with UQ policy.
7. Feel free to use the code/techniques you learned during your tutorial.
8. Please read the *How you may use generative AI in assessment*.

The aim is not to get a perfect answer, but to show that you know how to approach the problem and can describe the image analysis challenges, therefore your assignment needs considerable focus on method selection, algorithms, and discussion of the results obtained. The assignment should be presented in the form of a brief report including code, images, and descriptive text.

Task1: Create a video collage – 5 marks

Take a video's frames as input and create a collage of frames of that video. Please select video by your own.



Figure 1: source: <https://www.procollage.com/styles/blended-photo-collages.php>

Chose five frames from the video *randomly* and join them into a collage. Sort the frames based on their colour/edge information. An example for image placement in collage: image with less variation in colour can be in the centre and the ones on the boundary of the collage have more colour variation.

1. *CollageCreate(AddressofFolder)* function in *CollageCreator.py* returns and displays a collage image. The input to the function is a relative address of folder containing images from a video.
2. In the word document *Report.doc* mention the method in *CollageCreate()*. How are the images joined together.

Hint: Histograms, colour and edge information can be used. For extracting frames from a video, you can use *ffmpeg* library.

Task 2: Road sign Detection – 5 Marks

Modern vehicles (e.g. [Waymo](#)) increasingly rely on computer vision systems that can detect and interpret street signs to provide speed-limit alerts and other road information.



Figure 2: Image source: Alamy

Your task is to automatically identify and localize street signs in the provided image set (see *StreetSigns.zip* on Ultra [Learn.UQ]). You may begin by performing 2D cross-correlation using a template of the sign's border, but you will likely need to “chamfer” the template by smoothing it with a Gaussian or similar blur. A strong edge-detection method, such as the Canny Edge Detector, will also be essential. Apply your approach to the additional sample images in *StreetSigns.zip* and present your results.

In your report, discuss methods used, problems encountered, performance, and possible solutions. Comment on the problems encountered in sign extraction and the difficulties in designing a general sign detector.

Hint: http://www.gavrila.net/Research/Chamfer_System/chamfer_system.html.

Task 3: Cartoonize Face – 10 Marks

our input image contains a single face. Convert the image into a cartoon. The face (and body) should be cartoonized. The background should be blurred (for simplicity no need to cartoonize the background). There can be various ways for cartoonizing like k-means based colour quantization. However, the results may not be very convincing around the boundary. Another possible solution to use bilateral filtering (read it!).

The function name is *CartoonNizer(image)*. It takes the input image as parameter and displays and saves the cartoonized output.

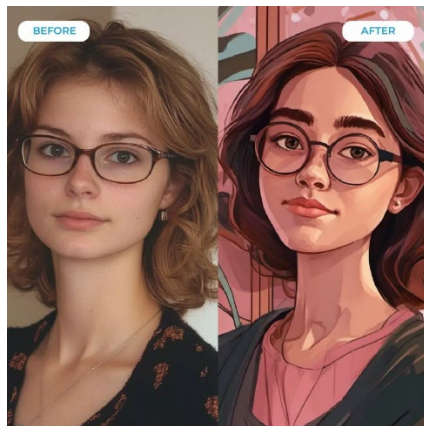


Figure 3: Image source
<https://pixlr.com/design/photo-to-cartoon/>

Implement the following three variants for completing the task above –

1. Your own K-means clustering based implementation.
2. Bilateral filtering-based implementation.
3. Improved (your own/inspired) method and implementation (using the methods you learned during your tutorial Canny edge detector, skin segmentation, interest points, dilation, erosion etc.)

Hint: Think from skin segmentation perspective (you learned in your tutorial) and applying computer vision on top of segmented part!

Bilateral filtering:

http://homepages.inf.ed.ac.uk/rbf/CVonline/LOCAL_COPIES/MANDUCHI1/Bilateral_Filtering.html

Report containing details of your approach. In your report, discuss methods used, problems encountered, performance, and possible solutions.

More assignment instructions:

During the assignment marking the file *CollageCreator.py*, *SignRecognition.py* and *CartoonNizer.py* will be executed directly and the outputs will be checked.

No change is going to be made to the code at our end. The image folder (for the tasks above) should be part of the zip file, which you will upload on Ultra (Learn.uq). In the image folder do not store a lot of images. Make sure that the resolution of each image is $\leq 640 \times 480$.

Please feel free to use any python interface you are comfortable with. For example: notebook i.e “.ipynb” file (Interactive PYthon NoteBook in Jupyter or VScode) where you can add notes in between your codes; or “.py” file which is widely used for python codes.

For report writing, please focus on explanation rather than writing theoretical background. We want to see if you can apply the computer vision algorithms to interesting real-world problems! For this part, we will conduct face to face 5 min interview. We will share details of interview after the submission deadline.

Assignment 1 Marking Scheme and Criteria

Q1

- Coding of the solution to generate and display collage image. Please add one/two liner comments in the code (similar to your tutorial codes). (3 marks)
- Describe how the collage image is generated in the report. Comment on the many challenges (lighting, day night vision, rain, clouds or other factors present in the video/image frames) and suggest possible solutions. Please do not add code snippet/screenshot in the report part. Here, we want to see your understanding of logic. For this part, we will conduct face to face interview. We will share details after the submission deadline. (2 marks)

Q2

- Coding of a solution to locate one sign, description and explanation of method, related images and graphs. Please add one/two liner comments in the code (similar to your tutorial codes). (2 marks)
- Testing method on all other signs, modification of method as required, showing related images with detection overlay. (1 marks)
- Comment on the many challenges to road sign detection including lighting, different shaped signs, different coloured signs, perspective distortion, angle of rotation, and so on and suggest possible solutions. Please do not add code snippet/screenshot in the report part. Here, we want to see your understanding of logic. For this part, we will conduct face to face interview. We will share details after the submission deadline. (2 marks)

Q3

- Coding of K-means clustering based implementation. Please add one/two liner comments in the code (similar to your tutorial codes) (1 mark)
- Coding of Bilateral filtering-based implementation. Please add one/two liner comments in the code (similar to your tutorial codes) (2 marks)
- Improved (your own/inspired) method and implementation. Please add one/two liner comments in the code (similar to your tutorial codes) (3 marks)
- A detailed report mentioning K-means clustering, Bilateral filtering-based, and your own improvements with output differences. Comment on the many challenges and suggest possible solutions. Please do not add code snippet/screenshot in the report part. Here, we want to see your understanding of logic. For this part, we will conduct face to face interview. We will share details after the submission deadline. (4 marks)